INCH-POUND
MIL-C-24689B
23 April 1992
SUPERSEDING
DOD-C-24689A
11 August 1989
(See 6.7)

MILITARY SPECIFICATION

# COBALT ALLOY CASTINGS, WEAR AND CORROSION RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers castings of cobalt-chromium-tungsten and cobalt-chromium-nickel alloys for use where corrosion and wear resistant materials are required.

1.2 Classification. Castings are of the following compositions, as specified (see 6.2).

(a) Composition No. 20 (UNS R30006).
(b) Composition No. 21.
(c) Composition No. 22 (UNS R30001).
(d) Composition No. 23 (UNS R30021).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-C-3993 - Copper and Copper-Base Alloy Mill Products; Packaging of.

STANDARDS

MILITARY

MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-792 - Identification Marking Requirements for Special Purpose Components.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials. (DoD adopted)
E 354 - Standard Methods for Chemical Analysis of High Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. Material shall be made by an electric furnace process. Unless a specific type is specified (see 6.2), static, centrifugal, or investment castings may be supplied at the option of the contractor.



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3.3 Process prohibitions. Unspecified heat treatments, cold working, straightening, or weld repair shall not be permitted.

3.4 Descaling. Castings shall be cleaned so that the surface is free from scale and foreign matter. If shot, grit, or vapor blasting is used, it shall be followed by machining, grinding, or pickling (provided a minimum of 0.10 inch (0.25 millimeter) is removed in subsequent machining) when liquid penetrant inspection is required.

3.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.6 Chemical composition. The casting material shall conform to the chemical composition specified in table I. Data may be ordered to verify this requirement (see 6.3).

TABLE I. Chemical composition.

| Element | Analysis (percent) 1/ | | | |
|---|---|---|---|---|
| | Composition number | | | |
| | 20 | 21 | 22 | 23 |
| Carbon | 0.90-1.4 | 1.8-2.2 | 2.0-2.6 | 0.20-0.30 |
| Manganese | 1.0 | 0.50 | 1.0 | 1.0 |
| Silicon | 1.5 | 0.80-1.5 | 1.5 | 1.0 |
| Chromium | 27.0-31.0 | 23.0-25.0 | 29.0-33.0 | 25.0-29.0 |
| Tungsten | 3.5-5.5 | 11.0-13.0 | 11.0-14.0 | ---- |
| Nickel | 3.0 | 3.0 | 3.0 | 1.75-3.75 |
| Iron | 3.0 | 2.0 | 3.0 | 3.0 |
| Molybdenum | 1.5 | 1.0 | 1.0 | 5.0-6.0 |
| Others, total (Ti, As, P, S, Cu, Mo, B, Zn) | 1.0 | 1.0 | 1.0 | ---- |
| Phosphorus | ---- | ---- | ---- | 0.04 |
| Sulfur | ---- | ---- | ---- | 0.04 |
| Boron | ---- | ---- | ---- | 0.007 |
| Cobalt | Remainder | Remainder | Remainder | Remainder |

1/ Maximum unless a range is shown.



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3.7 Mechanical properties. The material shall conform to the mechanical properties specified in table II. Data may be ordered to verify this requirement (see 6.3).

TABLE II. Hardness (Rockwell "C").

| Alloy | Static and centrifugal castings 1/ | Investment castings 1/ |
|---|---|---|
| Composition 20 | 40 | 37 |
| Composition 21 | 50 | 48 |
| Composition 22 | 50 | 48 |
| Composition 23 | 23-34 | 23-34 |

1/ All values minimum except where a range is shown.

3.8 Heat treatment. Composition 23 castings shall not be stress relieved. Composition 20, 21, 22 castings shall be stress relieved at 1650 ± 25 degrees Fahrenheit (°F) (900 ± 14 degrees Celsius (°C)) for 1 hour per inch of cross section with 2 hours minimum total time at temperature. Castings shall be cooled in the furnace at 250°F (120°C) per hour maximum to 500°F (260°C). Below 500°F (260°C) the castings may be cooled in still air to ambient temperature. Cooling may be done in steps provided the specified cooling rate is maintained.

3.9 Soundness. Castings shall be of uniform quality and condition, free of cracks, burrs, sand, tears, shrinkage, cold shuts, excessive scale, fins, porosity, and segregation to the extent they meet the visual and nondestructive inspection requirements.

3.10 Removal of surface defects. Surface defects may be ground out provided the ground out area is radiused and does not reduce the castings dimensions below the minimum dimensions of the casting drawing. The bottom radius of the ground out area shall be at least three times its depth.

3.11 Weld repair. Weld repair is not permitted without specific approval from the command or agency concerned. When weld repair is approved, the welding procedure will be approved as directed by the repair approval authority.

3.12 Dimensions. Castings shall be so that they can be laid out and machined to the finished dimensions of the casting drawing, within the specified tolerances. Sufficient stock shall be allowed for finishing.

3.13 Identification. Each casting shall be permanently marked as specified in 3.13.1 through 3.13.3.

3.13.1 Information required. Individual castings shall be marked with the following:

(a) Foundry's name or trademark.
(b) Specification and composition number.
(c) Pattern or piece number.
(d) Heat number and heat treat lot number.



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3.13.2 Methods and location of marking. The methods of marking shall be permanent marking and in accordance with MIL-STD-792. The location of marking shall be at a low stress area as shown on the drawing or on an area that will not be removed by machining.

3.13.3 Small castings. Where individual castings cannot be marked as specified in 3.13.1 and 3.13.2, they may be bundled or boxed and each bundle or box marked with a metal tag containing the specified information.

3.14 Alloy identity. Prior to shipment, all castings shall be verified to be cobalt-chromium-molybdenum-nickel alloy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the test as specified in 4.3.2.

4.3.1 Sampling for first article inspection. The first castings submitted for inspection shall be the first article sample.

4.3.2 Radiographic inspection. The first article casting shall be radiographically inspected in accordance with MIL-STD-271 to the acceptance criteria in accordance with MIL-STD-278 of the criticality level specified, or to another standard, as specified (see 6.2).



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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests as specified in 4.5 and 4.6.

4.4.1 Lot definition. Lots shall be as defined in 4.4.1.1 through 4.4.1.2.

4.4.1.1 Lot for chemical analysis. Lot for chemical analysis shall be as follows:

4.4.1.1.1 Master melt used. When a master melt is used, a lot shall consist of all material cast from the same master melt, by the same process, and presented for inspection at the same time.

4.4.1.1.2 Master melt not used. When a master melt is not used, a lot shall consist of all the castings cast from the same heat, by the same process, and presented for inspection at the same time.

4.4.1.1.3 Carbon additions. When carbon additions are made during pouring of castings and for the purpose of carbon analysis only, a lot shall be further defined as the castings poured between carbon additions.

4.4.1.2 Visual and dimensional examination, and mechanical property tests. A lot shall consist of all the castings from a lot as defined in 4.4.1.1.1 or 4.4.1.1.2 that have the same maximum cross-section and which have been heat treated together.

4.4.2 Sampling. Sampling for quality conformance inspection shall be as specified in 4.4.2.1 through 4.4.2.4.

4.4.2.1 Sampling for visual and dimensional examination. Each casting shall be examined visually and dimensionally.

4.4.2.2 Sampling for chemical analysis. Two samples shall be selected at random from each lot. Samples may be taken from the ladle or from casting blanks. When taken from the ladle, the samples shall be poured at the same time the castings are poured. When carbon additions are made to the melt during pouring of the castings, samples shall be taken for carbon analysis as follows: one immediately before each addition and one after each addition, following an appropriate hold time for homogenization.

4.4.2.3 Sampling for hardness tests. Two castings shall be tested from each lot.

4.4.2.4 Sampling for nondestructive inspection. Each casting shall be inspected.

4.5 Visual and dimensional examination. Castings shall be examined for conformance to 3.4, 3.9, 3.12, 3.13 and the criteria below. Any casting which fails to meet the dimensional requirement of the casting drawing shall be rejected. The criteria for acceptance of visual surface porosity shall be as follows: Critical machined surfaces, as specified (see 6.2) shall be free of porosity. For all other surfaces, porosity with a maximum opening of 3/32 inch or less is acceptable provided that the internal surfaces are visible, the depth does



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not exceed 1/8 inch, and the minimum wall thickness (determined from drawing dimensions) is not violated. Surface porosity with greater than 3/32 inch opening, or deeper than 1/8 inch, or having internal surfaces that are not visible, shall be explored by grinding (see 3.10) or the casting shall be rejected.

4.6 Test methods.

4.6.1 Chemical analysis. Chemical analysis shall be performed by wet or spectrochemical methods as determined by the contractor. In case of dispute, the wet method as specified in ASTM E 354 shall govern. Chemical composition shall be as specified in 3.6.

4.6.2 Hardness test. Hardness tests shall be performed in accordance with ASTM E 18. Hardness shall be as specified in 3.7.

4.6.3 Nondestructive testing. When specified on the casting drawing, castings shall be liquid penetrant inspected or radiographed to the extent specified. Test method shall be in accordance with MIL-STD-271. Acceptance criteria shall be as specified on the casting drawing or in the contract or purchase order. Data may be ordered to verify this requirement (see 6.3).

4.7 Inspection of packaging. Sample packages, and the inspection of the packaging, (packaging, packing and marking) for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packing requirements. Packaging (packaging, packing and marking) levels and other packaging ordering data options shall be in accordance with MIL-C-3993. Unless otherwise specified (see 6.2), level C packaging and level C packing shall apply. When specified (see 6.2) level A packaging and level A or B packing as specified (see 6.2) shall apply. In addition, for Navy acquisition, the following shall apply:

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer materials used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated materials conforming to MIL-L-19140 as follows:

Level A and B - Type II - weather resistant.
Category I - general use.
Level C - Type I - non-weather resistant.
Category I - general use.



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(2) Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class-domestic/fire retardant or class-weather resistant/fire retardant materials equipment as specified (see 6.2), of PPP-F-320 and amendments thereto.

(3) Cushioning and wrapping materials. The use of excelsior, newspaper, shredded paper (all types), and similar hydroscopic or non-neutral materials and all types of loose fill materials for packaging applications such as cushioning, fill, stuffing and dunnage is prohibited. Materials selected for cushioning and wrapping shall have properties (characteristics) for resistance to fire (see 6.7). Cushioning or wrapping materials, as applicable, shall be provided to prevent item and package damage and to prevent free movement of the container contents.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These castings, with high abrasion resistance, excellent corrosion resistance, and high hardness, are intended for use as bearings, valve discs, bushings, and seals, operating under high loads with little or no lubrication. Alloy R30001 has higher abrasion resistance than alloy R30006, but with lower impact strength.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Composition required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) If first article inspection is required (see 3.1).
(e) Type of casting, if a specific type is required (see 3.2).
(f) First article radiographic standard or criticality level, if MIL-STD-278 is invoked (see 4.3.2).
(g) When level C packaging and level C packing will not apply (see 5.1).
(h) When level A packaging applies; level of packing required, and other packaging ordering data requirements required (see 5.1).
(i) When fire-retardant lumber and plywood is required (see 5.1(a)(1)).
(j) Class of fire retardant fiberboard required (see 5.1(a)(2)).



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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 3.6 and 3.7 | DI-MISC-80705 | Certification data for non-nuclear level 1 material | ---- |
| 4.6.3 | DI-MISC-80678 | Certification/data report | 10.3.1 does not apply |

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 Basic foundry practices. The contractor should maintain a record of the basic foundry practices (type of melting, and refining units, molding process, basic rigging design, location of gates, risers, chills) used in the first article casting and make no change thereto without the specific approval of the command or agency concerned. The contractor may be required to perform specific first article tests and examinations to verify that any change will not degrade casting quality.

6.5 Replacement, retest, rejection and resubmittal.

6.5.1 Replacement of test specimens. A test specimen (for chemical analysis) or location (for hardness) may be discarded and a replacement test specimen selected from the same lot of material in accordance with FED-STD-151.



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6.5.2 Retest. Retesting will be permitted for chemical analysis and hardness tests, as follows:

(a) When one or more specified characteristic fails to conform to the requirements of this specification, retest of the nonconforming characteristic is permitted to establish product acceptability.
(b) Retesting should be performed on twice the number of representative specimens that were originally nonconforming.
(c) Unless otherwise approved by the acquisition activity, retest specimens should be taken adjacent to the initial nonconforming test specimen location.

6.5.3 Rejection. When any retest specimen does not conform to specification requirements for the characteristic being tested or all tests of a resample fails to meet the specified requirements for a particular property, the lot represented by the specimen should be rejected.

6.5.4 Resubmittal of rejected lots. A rejected lot may be resubmitted for acceptance testing only after rework, as necessary, is performed to correct the nonconforming condition without adversely affecting the conforming properties. If the rejected lot is re-heat treated to correct a nonconforming characteristic, all mechanical properties, including those which were initially conforming, should be determined.

6.5.5 Resubmittal of individual pieces. At the option and expense of the contractor, when a rejected lot consists of more than one piece, each remaining piece in the lot may be tested for the nonconforming characteristic and each piece that conforms to all specification requirements may be offered for acceptance.

6.6 Subject term (key word) listing.

Bearing
Bushing
Chromium
Master melt
Seals
Tungsten

6.7 Cushioning and wrapping materials (see 5.1(a)(3)). Materials having properties for resistance to fire and acceptable for use within interior (unit and intermediate) packs and shipping containers for Navy acquisitions are:



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| Material | Specification |
|---|---|
| Cushioning Material, Plastic, Open Cell | A-A-440 |
| Paper, Kraft, Treated (Fire Resistant) | A-A-1894 |
| Paper, Kraft, Wrapping | UU-P-268, Type II, Grade C or D |
| Fiberboard | PPP-F-320, see 5.1(a)(2) |
| Plastic Film, Flexible, Cellular | PPP-C-795, Class 3 - Fire Retardant |
| Polystyrene Expanded, Resilient | PPP-C-850, Grade SE |
| Plastic, Open Cell, Cushioning | PPP-C-1842, Type I, Style B |
| Bound Fiber | PPP-C-1120, Class A, Grade 1, Type Optional |
| Rubber, Latex Foam | MIL-R-5001, Grade A |
| Rubber, Cellular | MIL-R-6130, Grade A |
| Fibrous Glass | MIL-C-17435 |
| Polystyrene Foam | MIL-P-19644, Type II |
| Rubber, Cellular, Synthetic | MIL-R-20092, Class 5 |
| Polyurethane Foam | MIL-P-26514 |
| Cushioning, Resilient Type, General | MIL-C-26861 |
| Polyurethane Foam, Flexible, Open Cell | MIL-F-81334 |
| Foam-In-Place Packaging Materials: General Specification For | MIL-F-83671 |
| Foam, Combustion, Retardant, for Cushioning Supply Items Aboard Navy Ships | MIL-F-87090 (SA) |

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - SH
Air Force - 20

Review Activities:
Army - MI, AR, ME
Navy - AS
DSC - IS

Preparing activity:
Navy - SH
(Project MECA-0472)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-C-24689B | 930423 |

**3. DOCUMENT TITLE**

COBALT ALLOY CASTINGS, WEAR AND CORROSION RESISTANT

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| TECHNICAL POINT OF CONTACT:<br>MR. JOHN FORNEY, SEA 5141 | (1) Commercial<br>703-602-0205 | (2) AUTOVON<br>332-0205 |
| c. ADDRESS (Include Zip Code)<br>Commander, Naval Sea Systems Command<br>SEA 55Z3<br>Washington, DC 20362-5101 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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